

Prashant Kumar

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SUMMARY

M.Tech Control & Instrumentation student targeting Analog/Mixed-Signal and VLSI fresher roles, with Cadence Virtuoso schematic/layout exposure, DRC/LVS, PEX, post-layout simulation, MOSFET modeling, active filters, and transistor-level CMOS analog design.

EDUCATION

Delhi Technological University <i>M.Tech, Control & Instrumentation; 74.5%</i>	New Delhi 2024 – 2026
MJP Rohilkhand University <i>B.Tech, Electrical Engineering; 69.2%</i>	Bareilly 2017 – 2021

TECHNICAL SKILLS

Analog/MS IC: MOSFET operation, small-signal analysis, current mirrors, differential pairs, OTA/op-amp fundamentals, active filters, gm/ID, EKV concepts
EDA/Simulation: Cadence Virtuoso Schematic/Layout, Spectre/Assura flow exposure, OrCAD PSpice, MATLAB/Simulink
Verification/Layout: DRC, LVS, PEX, post-layout simulation, PVT/Monte Carlo analysis, transistor sizing, guard rings, matching-aware layout
Programming: Basic Python, TCL, Verilog, C++; Linux/Red Hat workflow; **Self-Study:** PLL basics, ADC/DAC fundamentals, high-speed transceiver concepts

RELEVANT PROJECTS

- VCII-Based Simulated Grounded Inductor** | *Cadence Virtuoso, 0.18 μ m CMOS* 2025
- Designed custom analog layout of a 12-transistor VCII circuit in Cadence Virtuoso; completed DRC/LVS and PEX for schematic-to-layout correlation.
 - Validated 2.5 mH equivalent inductance, 50 kHz–2.5 MHz bandwidth, and 0.65 mW power for active-filter application.
 - Analyzed transistor sizing, parasitic loading, bandwidth, and power tradeoffs in CMOS analog implementation.
- Two-Stage CMOS Operational Amplifier** | *Independent Analog Design Exercise, 180 nm CMOS* 2026
- Prepared Miller-compensated CMOS op-amp topology using differential input pair, current-mirror load, common-source gain stage, bias branch, and compensation capacitor.
 - Defined DC/AC/transient verification plan for operating point, gain, phase margin, GBW, slew rate, output swing, power, CMRR, PSRR, and load stability.
 - Mapped layout practices for matching/parasitic control: symmetric pair, common-centroid mirrors, dummy devices, guard rings, and short high-impedance routing.
- CMOS-Based Grounded Capacitance Multiplier using PFTFN** | *OrCAD PSpice, 0.35 μ m CMOS* 2025
- Designed 14-transistor cascode OTA-based capacitance multiplier achieving 1.503 nF equivalent capacitance with 0.2% accuracy.
- Single-OTRA Multifunctional Biquad Filter** | *M.Tech Thesis, Cadence/ngspice, 180nm CMOS* 2025–2026
- Verified biquad at behavioral, transistor, and layout-extracted levels; built ngspice+Pytest automated testbench for bias sweep and corner analysis.
 - Identified HPF passband droop root cause via systematic V_{B1} sweep; resolved to <1dB. All 3 responses validated against published targets.
- AnalogCheck – RAG-Enhanced SPICE Netlist Debugger** | *Python, ChromaDB, LLM, ngspice* 2026
- Built CLI tool detecting semantic errors in SPICE netlists via token-split parsing, deterministic checks (floating node, missing ground, M vs MEG), and LLM reasoning.
 - Implemented RAG pipeline indexing 20+ IC error patterns (LM741, LM358, LM317 pinout) and PSpice convergence docs into ChromaDB with sentence-transformer embeddings.
 - RAG context at check-time grounds LLM diagnosis in verified reference docs. 35 chunks, 35ms retrieval, integrated into CLI with `–rag` flag.

EXPERIENCE

- Teaching Assistant, Department of Electrical Engineering** Aug 2024 – Present
Delhi Technological University *New Delhi*
- Supported academic/laboratory activities while strengthening fundamentals in circuit analysis, control, instrumentation, and simulation.

ACHIEVEMENTS

- Qualified GATE 2024 in Electrical Engineering.
- Published IEEE conference paper: “Application & Performance Comparison of Electrical Motors in Hybrid Electric Vehicles.”